Types of Auxin

Natural	Synthetic
Auxin occuring in plants are called "Natural auxin"	These are synthesized artificially and have properties like Auxin.
1. Indole Acetic Acid (IAA)	1. 2,4-Dichloro Phenoxy Acetic Acid (2,4-D)
2. Indole Propionic Acid (IPA)	2. 2,4,5-Trichloro Phenoxy Acetic Acid (2,4,5-T)
3. Indole Butyric Acid (IBA)	3. Napthalene Acetic Acid (NAA)
4. Phenyl Acetic Acid (PAA)	

(i) Free auxin

They move out of tissues as they are easily diffusible. Example: IAA.

(ii) Bound Auxin

They are not diffusible. Example: IAA.

4. Precursor

The amino acid Tryptophan is the precursor of IAA and zinc is required for its synthesis.

5. Chemical structure

Auxin has similar chemical structure of IAA.

6. Transport in Plants

Auxin is polar in transport. It includes basipetal and acropetal transport. Basipetal means transport through phloem from shoot to root and acropetal means transport through xylem from root to shoot.

7. Bioassay (Avena Curvature Test / Went Experiment)

Bioassay means testing of substances for their activity in causing a growth response in a living plant or its part.

The procedure involves the following steps:

When the *Avena* seedlings have attained a height of 15 to 30 mm, about 1mm of the coleoptile tip is removed. This apical part is the source of natural auxin. The tip is now placed on agar blocks for few hours. During this period, the auxin diffuses out of these tips into the agar. The auxin containing agar block is now placed on one side of the decapitated stump of *Avena* coleoptile. The auxin from the agar blocks diffuses down through coleoptile along the side to which the auxin agar block is placed. An agar block without auxin is placed on another decapitated coleoptile. Within an hour, the coleoptiles with auxin agar block bends on the opposite side where the agar block is placed. This curvature can be measured (Figure 15.8).

8. Physiological Effects

- They promote cell elongation in stem and coleoptile.
- At higher concentrations auxins inhibit the elongation of roots but extermely lower concentrations promotes growth of root.
- Suppression of growth in lateral bud by apical bud due to auxin produced by apical bud is termed as **apical dominance**.
- Auxin prevents abscission.
- It is used to eradicate weeds. Example: 2,4-D and 2,4,5-T.
- Synthetic auxins are used in the formation of seedless fruits (Parthenocarpic fruit).
- It is used to break the dormancy in seeds.

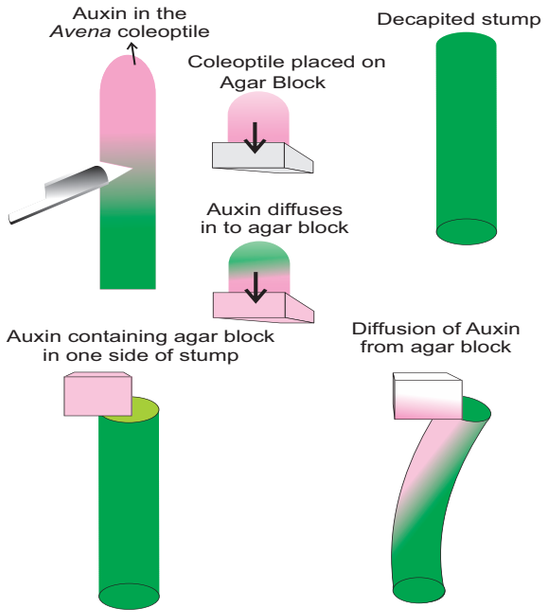


Figure 15.8: *Avena* Curvature Test

15.2.2 Gibberellins

1. Discovery

The effect of gibberellins had been known in Japan since early 1800 where certain rice plants were found to suffer from '**Bakanae**' or foolish seedling disease. This disease was found by **Kurosawa** (1926) to be caused by a fungus *Gibberella fujikuroi*. The active substance was separated from fungus and named as

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